

Nagar



Yuvak
Shikshan Sanstha's



Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to
Rashtrasant Tukadoji Maharaj Nagpur

University) Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website:

www.ycce.edu Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society. Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten) Nagar

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Session	2026-27 (ODD)	Course Name	Lab: JAVA STACK
Semester	5	Course Code	23CT1506
Roll No	64	Name of Student	Rushikesh Bhondekar

Practical Number	3
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Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	Perform File handling operations (read/write text files). Write a Java program to perform basic file-handling operations on a text file,
Problem Definition	such as creating, writing, reading, and appending data. File handling in Java is used to create, store, read, write, and modify data in files. Java provides several classes in the <code>java.io</code> package for performing file operations.
Theory (100 words)	A text file stores data in the form of characters. Java can perform different operations on text files using classes such as <code>File</code>, <code>FileWriter</code>, <code>FileReader</code>, and <code>BufferedReader</code>. Important File Handling Classes • <code>File</code> – Represents a file or directory and provides operations such as checking whether a file exists, creating a file, and obtaining file information. • <code>FileWriter</code> – Used to write character data into a text file. • <code>FileReader</code> – Used to read character data from a text file. • <code>BufferedReader</code> – Reads text efficiently and provides the <code>readLine()</code> method to read data line by line. File Opening Modes • Write mode ("w") – In Java, <code>new FileWriter("file.txt")</code> creates a new file or overwrites an existing file. • Append mode – <code>new FileWriter("file.txt", true)</code> adds new data at the end of the existing file without deleting previous data. • Read mode – <code>FileReader</code> is used to read data from the file. Basic File Operations 1. Create/Write: <code>FileWriter</code> is used to create a file and write data into it. 2. Read: <code>FileReader</code> and <code>BufferedReader</code> are used to retrieve and display the stored data. 3. Append: <code>FileWriter</code> with <code>true</code> as the second argument is used to add data at the end of the file.

4. Close: The `close()` method releases the file resource after the operation is completed.

5. Exception Handling: File operations may cause `IOException`, so they are generally handled using `try-catch`

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Procedure and Execution

(100 Words)

1. Create/open a text file using file-opening functions.
2. Write text into the file.
3. Close the file.
4. Reopen the file in read mode.
5. Read and display its contents.
6. Open the file in append mode and add new text.
7. Close the file and read it again to verify the changes.

Code:

```
import java.io.*;
```

```
public class FileHandling {  
    public static void main(String[] args) {  
        try {  
            // 1. Write data into the file  
            FileWriter fw = new FileWriter("sample.txt");  
  
            fw.write("Welcome to Java File Handling.\n");  
            fw.write("This is a text file.\n");  
        }  
    }  
}
```

```
fw.close();

// 2. Read data from the file
BufferedReader br = new BufferedReader(
new FileReader("sample.txt"));

System.out.println("File Contents:");

String line;
while ((line = br.readLine()) != null) {
    System.out.println(line);
}

br.close();

// 3. Append data to the file
FileWriter append = new FileWriter("sample.txt", true);

append.write("This line is appended to the file.\n");

append.close();

// 4. Read the updated file
br = new BufferedReader(
    new FileReader("sample.txt"));

System.out.println("\nUpdated File Contents:");

while ((line = br.readLine()) != null) {
    System.out.println(line);
}

br.close();

} catch (IOException e) {
    System.out.println("Error: " + e.getMessage());
}
}
```

```

import java.io.*;

public class FileHandling {
    public static void main(String[] args) {
        try {
            // 1. Write data into the file
            FileWriter fw = new FileWriter("sample.txt");

            fw.write("Welcome to Java File Handling.\n");
            fw.write("This is a text file.\n");

            fw.close();

            // 2. Read data from the file
            BufferedReader br = new BufferedReader(
                new FileReader("sample.txt"));

            System.out.println("File Contents:");

            String line;
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            br.close();

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            FileWriter append = new FileWriter("sample.txt", true);

            append.write("This line is appended to the file.\n");

            append.close();

            // 4. Read the updated file
            br = new BufferedReader(
                new FileReader("sample.txt"));

            System.out.println("\nUpdated File Contents:");

            while ((line = br.readLine()) != null) {
                System.out.println(line);
            }

            br.close();

        } catch (IOException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }
}

```

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Output:

File Contents:

Welcome to Java File Handling.

This is a text file.

Updated File Contents:

Welcome to Java File Handling.

This is a text file.

This line is appended to the file.

Output Analysis Link of student Github profile where lab assignment has been uploaded Conclusion	2. Detailed Line-by-Line Execution Analysis			
	<u>Execution Step</u>	<u>Class / Method Used</u>	<u>Operation Performed</u>	<u>Memory & System State</u>
	<u>1. Instantiation</u>	<u>new File("output.txt")</u>	<u>Creates an abstract path instance in memory.</u>	<u>No disk file created yet.</u>
	<u>2. Creation Check</u>	<u>file.createNewFile()</u>	<u>Interrogates the OS file system. Returns true if file is created; false if it exists.</u>	<u>Creates a 0-byte file on physical disk if not present.</u>
	<u>3. File Writing</u>	<u>new FileWriter(file)</u>	<u>Opens an output stream connected to <u>output.txt</u>.</u>	<u>Prepares destination for writing characters.</u>

<u>4. Buffer Flush</u>	<u>writer.close()</u>	<u>Flushes character stream to disk and frees the file lock.</u>	<u>Text is permanently written onto storage device.</u>
<u>5. File Reading</u>	<u>new BufferedReader(...)</u>	<u>Wraps FileReader to read chunked buffer blocks from disk into RAM.</u>	<u>Minimizes physical I/O overhead.</u>

<https://github.com/rushikesh-bhondekar/Java-lab->

[1. **Persistence:** Data was successfully written to secondary storage \(\[output.txt\]\(#\)\), allowing data retention beyond application execution.
2. **Stream Efficiency:** Combining \[FileReader\]\(#\) with \[BufferedReader\]\(#\) enabled memory-efficient line-by-line reading without loading the entire file into heap memory at once.
3. **Robustness & Resource Management:** Using \[try-catch\]\(#\) blocks effectively handles potential runtime \[IOException\]\(#\) instances \(e.g., permission failures or missing paths\), while automatic stream closures prevent system resource leaks.](https://github.com/rushikesh-bhondekar/Java-lab-Through this implementation, basic file I/O handling mechanisms in Java were successfully demonstrated:</p>
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Date	20/09/2026
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